

AI virtual-camera tracking that finds, identifies and holds the partner terminal's beacon, in software

❖ Proposed solution · Detailed explanation · How it addresses the problem · Innovation and uniqueness

PROBLEM EXISTING

- Micro-radian aim**
A hair-thin laser must stay on target while vehicles, drones and masts move. (Kaymak 2018)
- Wrong-target lock**
Stars, glints and lamps look like the beacon; brightness-only trackers lock the wrong light.
- Delay & dropouts**
≈40 ms command lag and blink gaps drop the beam, forcing slow manual re-alignment.

PROPOSED SOLUTION

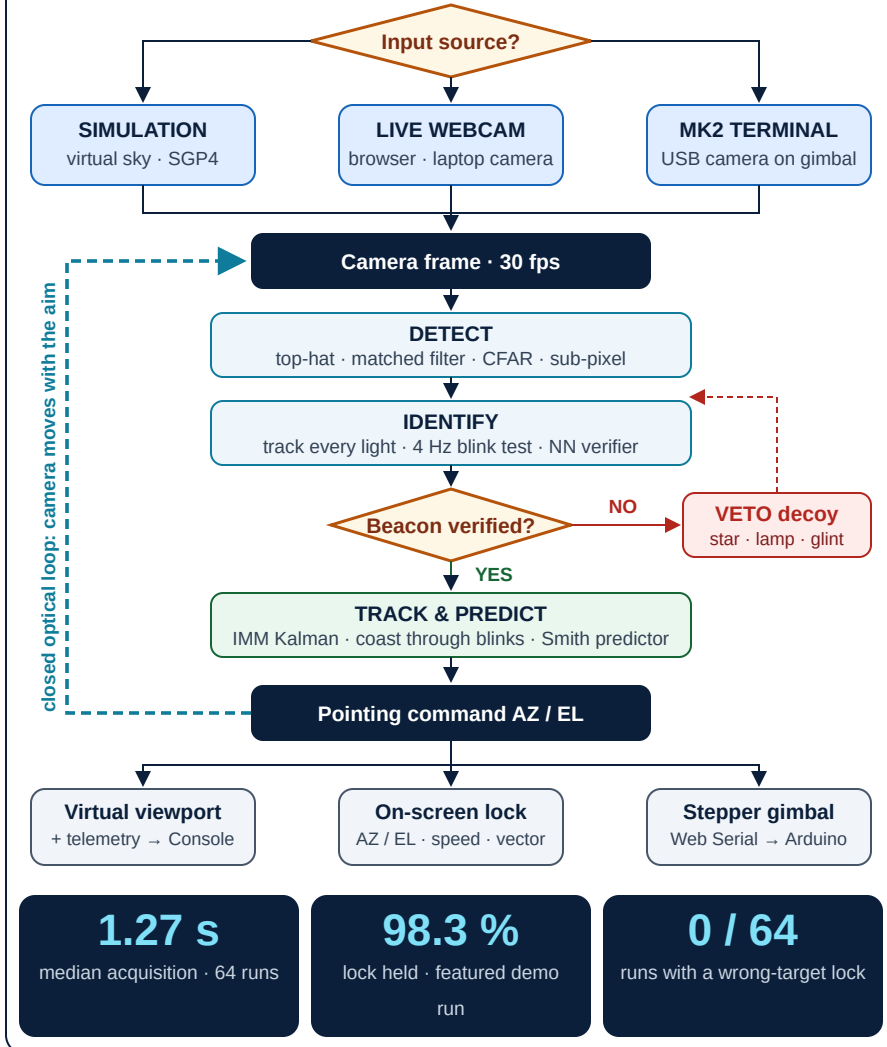
We propose **ZeroDrift**: software that finds the partner's **4 Hz beacon** in a camera image, **proves** it is the right light and **steers** the terminal to hold it centred: PS26169 solved without special hardware.

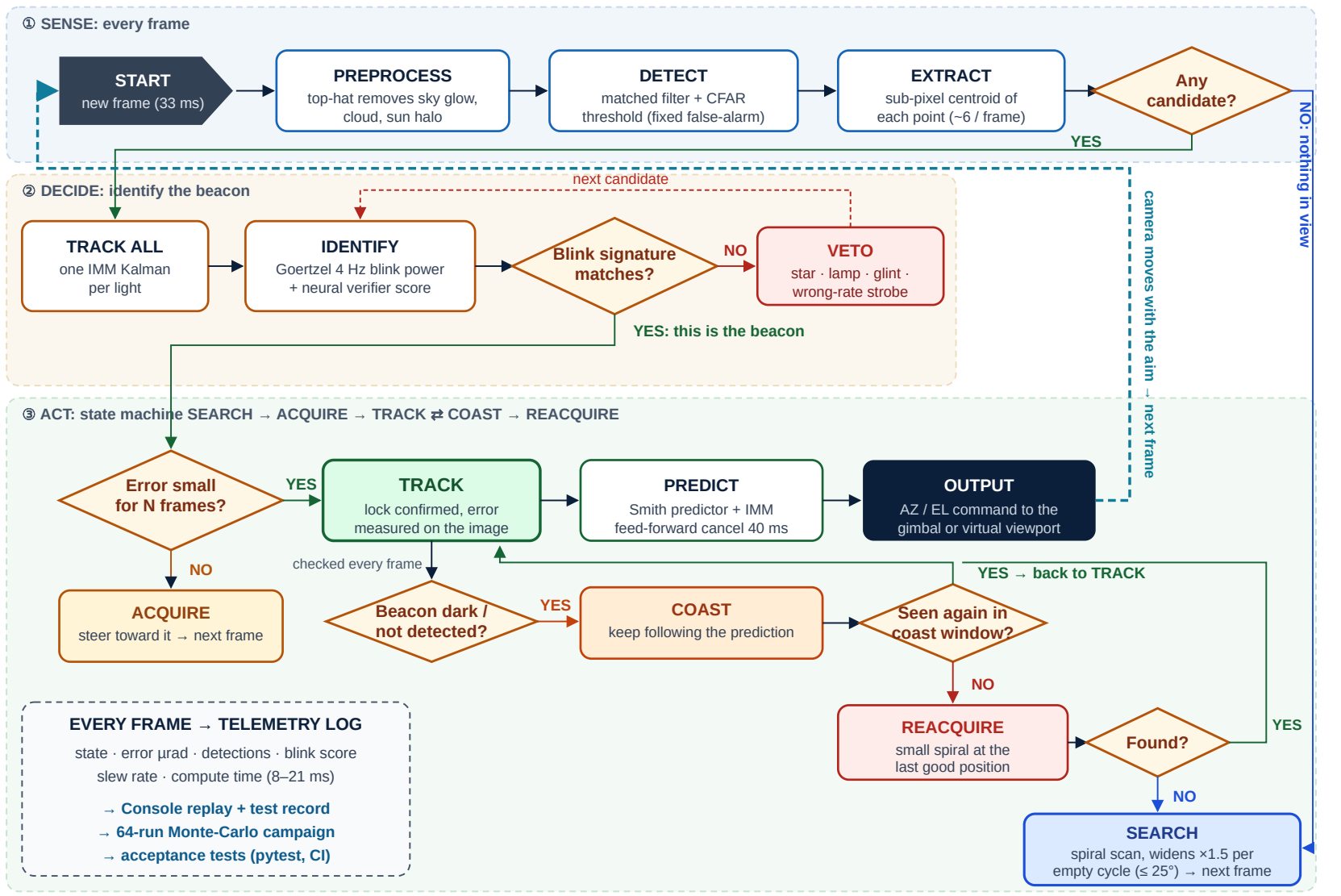
- Physics-true virtual camera**
SGP4 orbits, turbulence, decoys: **tested before hardware**
- Detect**
top-hat · matched filter · CFAR · **sub-pixel**
- Identify**
4 Hz blink test + neural verifier: **0 / 64 wrong locks**
- Track & coast**
IMM Kalman holds through blinks: **98.3 % held**
- Control**
Smith predictor cancels the command delay: **198 μ rad** median error, lock in **1.27 s**

UVP (UNIQUE VALUE PROPOSITION)

- Identity, not brightness**
Locks only on the blink signature
- Delay-cancelled loop**
Error measured on the image itself
- One engine, 3 modes**
Simulation · webcam · MK2 terminal
- Measured, not claimed**
Open source · 250 passing tests

ARCHITECTURE



SOFTWARE FLOWCHART · IMPLEMENTED IN src/fsoc_pat (pipeline · detection · tracking · control)

TECH STACK
ENGINE

Py Python 3.11 Np NumPy CV OpenCV Or SGP4 · ISS TLEs

Ym PyYAML scenes

AI / LOGIC

Cf CFAR detector Km IMM Kalman Hz Goertzel 4 Hz

NN Neural verifier Sp Smith predictor

APPS & WEB

Qt PySide6 GUI Pg PyQtGraph JS JavaScript

3D three.js twin Rt WebRTC camera Se Web Serial

HARDWARE

Ar Arduino C++ As AccelStepper St A4988 + NEMA17

I2 AS5600 · I²C

QUALITY & DEPLOY

Pt pytest · 250 pass CI GitHub Actions Ex PyInstaller app

Nf Netlify site

PROOF > PROMISE · WORKING PROTOTYPE


Console replay + test record

Live webcam lock, 4.0 Hz

MK1 rig, closed loop

1.27 s

median lock · 64 runs

0 / 64

wrong-target locks

198 μ rad

median error · demo

8–21 ms

per 33 ms frame · no GPU